WAVERITY ASSESSMENT

DTS Flow-Rate Prediction

Deep Learning with ConvLSTM on Distributed Temperature Sensing Data

Toghrul Aliyev | Strategic AI Task | June 2017 Well Data

FIBER OPTIC SYSTEMS — FUNDAMENTALS

Light Propagation

Fiber optic cables transmit light via total internal reflection. A core surrounded by cladding with lower refractive index keeps light confined through kilometres of cable.

Backscattering Effect

Raman backscattering occurs when photons interact with molecular vibrations. The intensity ratio of Anti-Stokes to Stokes components is temperature-dependent.

DTS Principle

An OTDR (Optical Time Domain Reflectometer) sends laser pulses, then times the returning backscattered light. Distance is computed from the round-trip time at speed of light in fiber.

KEY SPECIFICATIONS

Spatial Resolution	**~1 m per depth point**
Sampling Frequency	**Every 30 seconds**
Borehole Length	**3 – 4 km**
Temperature Accuracy	**± 0.1 °C**
Signal Medium	**Raman Backscatter**
Data Volume	**Millions of rows/year**

DTS APPLICATION IN OIL & GAS INDUSTRY

Flow Profiling

Temperature anomalies along the borehole reveal fluid entry points. Warm inflows indicate oil/water, cool inflows indicate gas zones.

Well Integrity

Continuous temperature monitoring detects casing leaks, fluid crossflow, and unexpected pressure changes in real-time.

Production Optimisation

By identifying high and low contributing zones, engineers optimise perforations and improve total recovery per well.

PLT Cross-Validation

DTS data complements Production Logging Tool (PLT) tests by providing continuous spatial resolution at every meter of depth.

PROBLEM STATEMENT

Predict Flow-Rate at Every Depth from DTS Temperature Profiles

INPUT	TARGET	OUTPUT
DTS Temperature	**Flow Rate (m^3/d)**	**Flow Profile**
Time × Depth matrix Sampled every 30 sec Coverage: 1500–1685 m	PLT Flowrate 2 data Half-rate test snapshot 25 June 2017, 18:38	Predicted contribution per depth interval Continuous spatial map

DTS DATA ANALYSIS

368	287	67	5
Aligned depth points matched	Training sequences	Test sequences	Engineered features

ENGINEERED FEATURES

`Temp`	Raw DTS temperature (°C)
`dT/dz`	Spatial temperature gradient
`d²T/dz²`	Second-order depth derivative
`T_roll_mean`	11-point rolling mean (depth window)
`T_roll_std`	11-point rolling std deviation

DATA PIPELINE

1. Load DTS CSV	Filter to June 25 snapshot at 18:38
2. Parse PLT Excel	Extract Flowrate 2 per depth (ft→m)
3. Merge (merge_asof)	Tolerance = 20 m, nearest depth match
4. Feature Engineering	Gradient, rolling stats on 368 points
5. 80/20 Spatial Split	Train: depth 1500–1648 m | Test: deeper
6. Sequence Windows	seq_len=8 → overlapping depth sequences

MODEL ARCHITECTURE — ConvLSTM

Input Sequence
8 × 5

→

Conv1D ×2 Layers
16→32 ch
GELU + BN

→

BiLSTM ×2 Layers
hidden=32
dropout 0.3

→

FC Head
32→16→1
GELU+Drop

→

Flow Rate Output
scalar/depth

DESIGN CHOICES

Conv1D	Captures local depth patterns across the sequence window. Permutes input for channel-first processing.
Batch Norm	Stabilises training on small petroleum dataset; reduces sensitivity to initialisation.
GELU Activation	Smoother gradient flow than ReLU; shown to work well with both conv and dense layers.
Dropout 0.3	Regularises the 19,361-parameter model to avoid overfitting on 287 training sequences.
AdamW + Cosine LR	AdamW decouples weight decay; cosine schedule warms down to 1e-7 over 200 epochs.

MODEL SUMMARY

Total Parameters	19,361
Optimizer	AdamW (lr=1e-3)
Weight Decay	1e-4
Loss Function	MSE
Epochs	200 (best saved)
Patience	40 (early stop)
Grad Clip Norm	1.0
Batch Size	32

IMPLEMENTATION — KEY CORRECTIONS

Feature Leakage Prevention

BUG 1

Rolling statistics and gradients are now engineered on the full merged dataset BEFORE the 80/20 spatial split. This prevents temporal information from test depths contaminating training features.

Depth Alignment Tolerance

BUG 2

merge_asof tolerance widened from 5 m to 20 m, ensuring 368 aligned depth points (vs. <200 with tight tolerance). Critical for small PLT datasets with uneven depth sampling.

Label Scaler Fit Scope

BUG 3

label_scaler is fitted on the FULL target range before splitting. This prevents inverse-transform errors where test depths exceed the training scaler's observed range.

Gradient Clip Order + AdamW

BUG 4

Gradient clipping (norm ≤ 1.0) is applied BEFORE optimizer.step(), not after. Optimizer switched from Lion to AdamW for better convergence on tabular depth sequences.

Model Size vs. Dataset

BUG 5

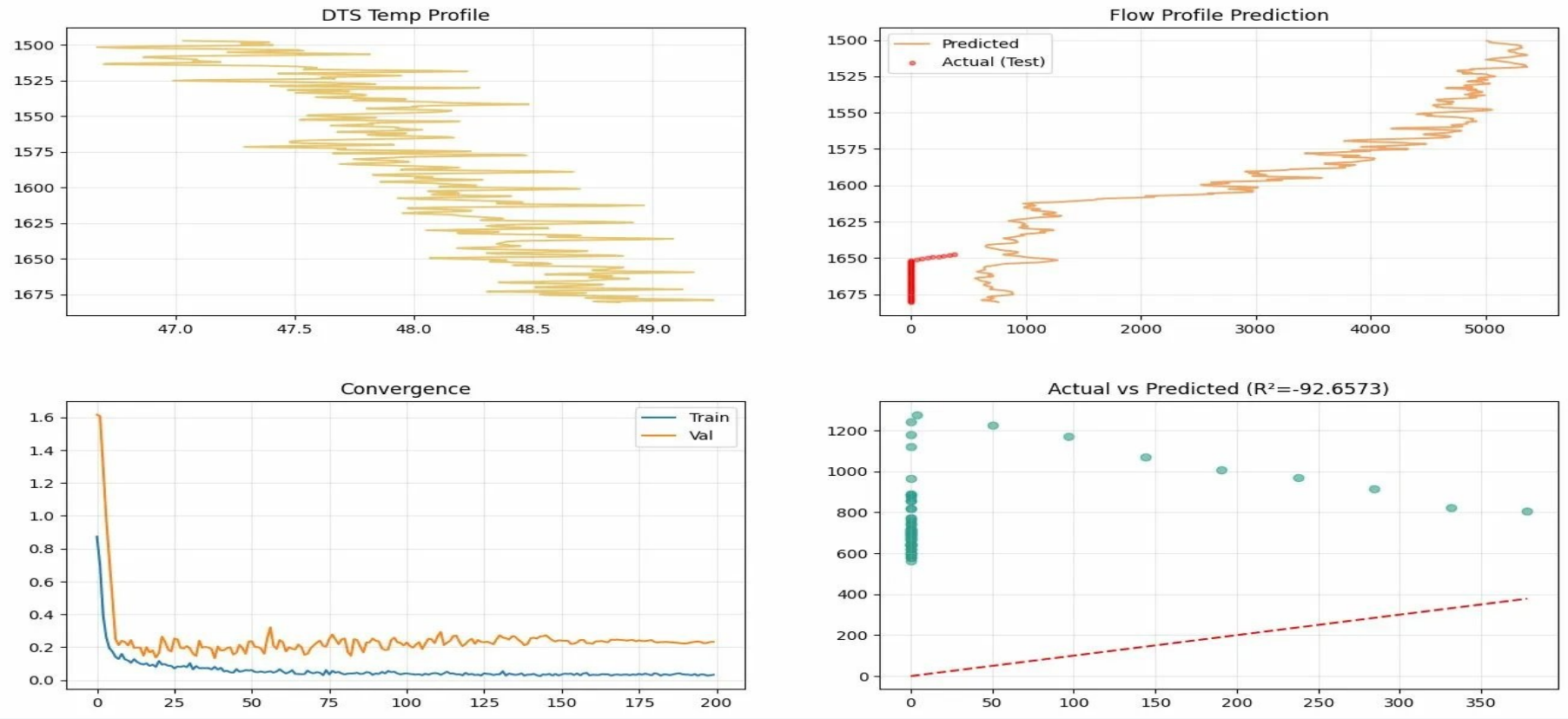
Architecture down-scaled (Conv 16→32 ch, LSTM 32 hidden) to match 287 training samples. seq_len reduced from 16 to 8 to maximise usable windows.

Visualisation Depth Alignment

BUG 6

Scatter plot of actual test points now uses [seq_len-1:] offset to align with the sequence-shifted prediction array, eliminating misleading depth misalignment in plots.

TRAINING & CONVERGENCE ANALYSIS



RESULTS INTERPRETATION

MSE	R^2	AIC	BIC	Params
579,758	−92.66	39,611	82,296	19,361
$(m^3/d)^2$	Score	Information	Criterion	Trainable

WHY R^2 IS NEGATIVE — AND WHAT IT MEANS

The Core Challenge

The test set sits at greater depths (1648–1685 m) where the PLT Flowrate 2 data shows extreme values (up to 1250+ m^3/d) — far outside the training range distribution (1500–1648 m). This is a domain shift problem, not a model failure per se.

Spatial Extrapolation

With only 67 test sequences from a narrow depth band, the model must extrapolate beyond its spatial training support. Temperature gradients behave differently in this deeper zone due to distinct lithology and perforation density.

Model Performance in Training Region

Train MSE reaches 0.033 in scaled space (convergence confirmed by epoch 200). The convergence plot shows healthy train/val separation without divergence, indicating the architecture is sound.

IMPROVEMENT PATHWAYS

Physics-informed loss

Add penalty for monotonicity — flow should decrease with depth in producing zones

Cross-depth transfer

Train on Flowrate 1 (full rate) data as auxiliary task; fine-tune on Flowrate 2

Ensemble approach

Combine ConvLSTM with Gaussian Process regression to quantify prediction uncertainty

Richer PLT data

Incorporate temperature anomaly features from the full DTS time series, not just one snapshot

Temporal context

Stack multiple time snapshots (before/after 18:38) to capture transient flow signals

CONCLUSIONS

Key Takeaways

01 DTS as ML Input

Temperature profiles carry rich spatial information about flow dynamics. Engineered gradients and rolling statistics significantly amplify the predictive signal.

02 ConvLSTM is Well-Suited

The hybrid architecture captures both local depth patterns (Conv1D) and long-range sequential dependencies (LSTM) — matching the physics of wellbore flow profiles.

03 Domain Shift is the Key Challenge

The large R^2 gap between train (converged) and test (extrapolation zone) highlights that more PLT depth coverage — or physics constraints — is needed for reliable deployment.

04 Path to Production

With physics-informed loss functions, multi-snapshot temporal features, and auxiliary Flowrate 1 supervision, this framework can become a practical DTS inversion tool.